

5.3.4

AI25BTECH11010 - Dhanush Kumar

Question

Find the product of the matrices $\begin{pmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ 3 & -3 & -4 \end{pmatrix} \begin{pmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{pmatrix}$ and hence solve the system of linear equations: $x + 2y - 3z = -4$

$$2x + 3y + 2z = 2$$

$$3x - 3y - 4z = 11$$

Solution

The given matrices are

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 3 & 2 \\ -3 & 2 & -4 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -6 & 17 & 13 \\ 14 & 5 & -8 \\ -15 & 9 & -1 \end{pmatrix} \quad (1)$$

The product of the matrices is computed as

$$\mathbf{AB} = \begin{pmatrix} 67 & 0 & 0 \\ 0 & 67 & 0 \\ 0 & 0 & 67 \end{pmatrix} = 67\mathbf{I}_3 \quad (2)$$

The given system of linear equations can be written in matrix form as

$$\mathbf{AX} = \mathbf{C}, \quad (3)$$

$$\mathbf{C} = \begin{pmatrix} -4 \\ 2 \\ 1 \end{pmatrix} \quad (4)$$

Since $\mathbf{AB} = 67\mathbf{I}_3$, the inverse of $\mathbf{A}$ exists and is given by

$$\mathbf{A}^{-1} = \frac{1}{67}\mathbf{B} \quad (5)$$

Multiplying both sides of $\mathbf{AX} = \mathbf{C}$ by $\mathbf{A}^{-1}$, we obtain

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{C} = \frac{1}{67}\mathbf{BC} \quad (6)$$

$$\mathbf{BC} = \begin{pmatrix} 201 \\ -134 \\ 67 \end{pmatrix} \quad (7)$$

Therefore, the solution of the system is

$$\mathbf{X} = \frac{1}{67} \begin{pmatrix} 201 \\ -134 \\ 67 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} \quad (8)$$

(9)